AEQUITAS NODE OPERATOR GUIDE

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Run a validator on your own server · Docker Compose · about 15 minutes, 10 of them building

Before You Start — What You Need

1. You are a registered human: Install the Aequitas app, complete the biometric registration, and note your wallet address. The network refuses a validator whose wallet is not a registered human — one human, one validator. Renting servers buys no extra vote.
2. A server (VPS) with a public IPv4 address: Ubuntu 22.04 or 24.04, at least 4 vCPU, 8 GB RAM, 60 GB SSD (the database is about 18 GB today and grows). The two founder nodes run on 6 vCPU / 12 GB / 100 GB. Ports 8080 (API) and 4001 (P2P) must be reachable from the internet.
3. Docker with the Compose plugin, and git. On Ubuntu: curl -fsSL https://get.docker.com | sh installs both.
4. About 15 minutes. Ten of them are the first build (the node is compiled from source on your server). Later updates take the same.

Step 1 — Configuration (.env)

Security warning: RELAYER_PRIVATE_KEY and NODE_KEY are secrets. Whoever has them IS your node. Never paste them into chat, e-mail or a ticket. The .env file stays on your server.

Variable	Required?	What to set
POSTGRES_PASSWORD	**YES**	A long random password for the local database. Used only by the two containers on your server.
SELF_URL	**YES**	How other nodes reach YOUR node: http://YOUR-PUBLIC-IP:8080 (or https://your-domain if you put a proxy in front). Without it the node only follows the chain as an observer and never registers as a validator.
NODE_OPERATOR_WALLET	**YES**	Your own wallet address — MUST be a registered human (app registration completed). This is what ties the validator to a person. Receives the validator rewards.
RELAYER_PRIVATE_KEY	**Recommended**	The key that signs your blocks (0x…, 66 characters). Simplest: the private key of NODE_OPERATOR_WALLET — then no extra binding is needed. Left empty, the node generates one on first start and prints it ONCE ("SAVE THIS AS …"); put it into .env and restart, or your identity changes on every restart.
NODE_OPERATOR_BINDING_SIGNATURE	**Only if the keys differ**	If RELAYER_PRIVATE_KEY is not the key of NODE_OPERATOR_WALLET: sign the message shown at aequitas.digital/node-binding with your wallet and paste the signature here.
NODE_KEY	**Recommended**	P2P identity. Generated and printed on first start if empty — save it into .env, same reason as above.
PRIMARY_NODE_URLS	**Preset**	The nodes yours registers with and fetches the state snapshot from on first start. Preset to the two founder nodes; leave as is.
GOMEMLIMIT	**Preset**	Memory ceiling of the node process, preset 5GiB (fits 12 GB RAM). On 8 GB RAM set 3GiB.

Variable	Required?	What to set
POSTGRES_SHARED_BUFFERS	**Preset**	Postgres cache, preset 1GB. A quarter of your RAM is a good value.

Step 2 — Start the node (Docker Compose)

Everything the two founder nodes do, as one file. The first command builds the node from source (about 10 minutes), starts Postgres and the node, and restarts both automatically after a crash or a reboot.

1. On your server: git clone https://github.com/hanoi96international-gif/Aequitas.git
2. cd Aequitas/deploy/validator and cp .env.example .env
3. Edit .env (for example with nano .env): fill in POSTGRES_PASSWORD, SELF_URL and NODE_OPERATOR_WALLET — see the table above
4. docker compose up -d --build — builds and starts. Takes about 10 minutes the first time
5. Watch the log: docker compose logs -f node. On first start the node imports the network state (snapshot) from a founder node, checks its signature, then pulls the blocks since then: [BOOTSTRAP] Fresh node — importing state from … followed by [HTTP-SYNC] Added … new blocks
6. If the log printed SAVE THIS AS NODE_KEY or SET THIS AS RELAYER_PRIVATE_KEY: copy those values into .env now and run docker compose up -d again — otherwise your node gets a new identity on every restart
7. Once caught up you will see [Block #…] lines — those are blocks YOUR node produced. Until then the node deliberately produces nothing (Frischer Knoten: … produziert nichts, bis er aufgeholt hat) — a node that has never seen the chain must not invent one

```
# deploy/validator/.env — the three required values
POSTGRES_PASSWORD      = a-long-random-password
SELF_URL               = http://YOUR-PUBLIC-IP:8080
NODE_OPERATOR_WALLET   = 0xYOUR_HUMAN_WALLET
# recommended: the key that signs your blocks (or leave empty on first start)
RELAYER_PRIVATE_KEY    = 0xYOUR_PRIVATE_KEY
# preset, leave as is
PRIMARY_NODE_URLS      = http://173.249.37.118:8080,http://194.163.188.71:8080
```

Step 2b — Update, restart, stop

The node keeps its state in two Docker volumes (database and transfer log). Updating rebuilds the image from the latest code; the state stays. Never restart every validator of the network at the same time — one after the other.

```
# Update to the latest code (about 10 minutes)
cd Aequitas && git pull && cd deploy/validator && docker compose up -d --build

# Restart only the node
docker compose restart node

# Stop everything (state is kept)
docker compose down

# Watch resource use
docker stats
```

Step 3 — Verify Your Node is Running

Compare your node with the network. Replace YOUR-PUBLIC-IP with your server address.

```
curl -s http://YOUR-PUBLIC-IP:8080/api/status | grep -oE '"height":[0-9]+'
curl -s https://aequitas.digital/api/status  | grep -oE '"height":[0-9]+'
→ Both numbers must be equal within a few blocks.

http://YOUR-PUBLIC-IP:8080/          → your own explorer
http://YOUR-PUBLIC-IP:8080/api/health/combined → everything the node measures about itself
```

Right after the first start the height is far below the network while the snapshot imports and the recent blocks are pulled. If it stays far behind for more than 15 minutes, check the log for [BOOTSTRAP] errors and that ports 8080 and 4001 are open.

Step 3b — Bind your wallet to the node key (only if they differ)

If RELAYER_PRIVATE_KEY is not the private key of NODE_OPERATOR_WALLET, prove once that both belong to you: open the page below, sign the shown message with your wallet, and put the signature into .env as NODE_OPERATOR_BINDING_SIGNATURE.

```
https://aequitas.digital/node-binding
```

With the simple single-key setup (RELAYER_PRIVATE_KEY = your wallet's key) this step is not needed.

Step 4 — Connect MetaMask to Your Node (Optional)

In MetaMask: network dropdown → Add network → Add a network manually, then enter:

Network Name	Aequitas Chain
RPC URL	http://YOUR-PUBLIC-IP:8080/rpc
Chain ID	1926
Currency Symbol	AEQ
Decimals	18
Block Explorer	https://aequitas.digital

Step 5 — Validator Rewards

The validators pool collects 40% of all protocol fees (swap fees, demurrage, wealth-cap overflow). Every day at 20:00 Berlin time the pool is distributed to the registered validators in proportion to the blocks they produced. Nothing to do beyond keeping the node running.

1. NODE_OPERATOR_WALLET must be a registered human — otherwise the network rejects the registration (log: NODE_OPERATOR_WALLET is not a registered human).
2. Confirm in the log: [PEERS] Auto-authorized validator … (wallet: 0x…) on a founder node, and [Block #…] lines on yours.
3. A node that is down produces no blocks and therefore earns nothing for that time — restarts are harmless, the node catches up on its own.

4 What a validator does NOT do without extra software: accept new human registrations (those endpoints answer 503). Transfers, blocks and rewards work without it.

Troubleshooting

Symptom	Likely Cause	Solution
NODE_OPERATOR_WALLET is not a registered human	The wallet has not completed the app registration	Register in the app first, then restart the node.
operator_binding_signature missing or invalid	Signing key and wallet differ, no binding	Step 3b: sign at /node-binding and set NODE_OPERATOR_BINDING_SIGNATURE.
Frischer Knoten: … produziert nichts, bis er aufgeholt hat	Normal while catching up	Wait. Production starts once the node has completed clean sync cycles with the founder nodes.
SELF_URL not set — … Beobachter	SELF_URL missing in .env	Set SELF_URL to http://YOUR-PUBLIC-IP:8080 and run docker compose up -d.
Height stays far below the network	Snapshot import failed, or ports closed	Check the log for [BOOTSTRAP] lines; open TCP 8080 and 4001 inbound in the firewall / cloud security group.
Node restarts in a loop, "OOMKilled"	Not enough RAM	Lower GOMEMLIMIT in .env (3GiB on 8 GB RAM) or give the server more memory.
docker compose: build fails	No outbound internet during build	The build downloads Go modules; check DNS and outbound HTTPS on the server.